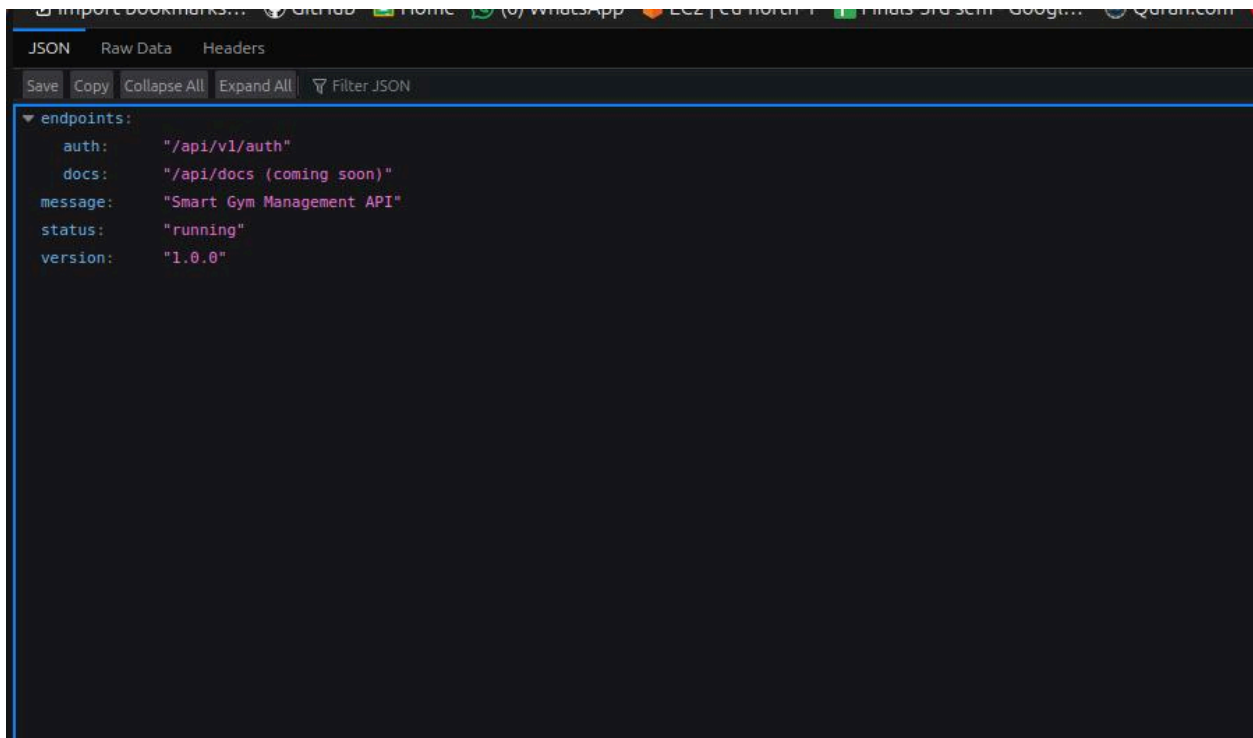




```
(venv) sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management/backend$ python3 run.py
✓ Database connection pool created successfully

🚀 Starting Smart Gym API on http://localhost:5000
📖 API Documentation: http://localhost:5000/api/docs (coming soon)
🏠 Health Check: http://localhost:5000/health
🔑 Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.100.11:5000
Press CTRL+C to quit
* Restarting with stat
✓ Database connection pool created successfully

🚀 Starting Smart Gym API on http://localhost:5000
📖 API Documentation: http://localhost:5000/api/docs (coming soon)
🏠 Health Check: http://localhost:5000/health
🔑 Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Debugger is active!
* Debugger PIN: 128-685-615
127.0.0.1 - - [06/Mar/2026 15:21:39] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [06/Mar/2026 15:21:39] "GET /favicon.ico HTTP/1.1" 404 -
```

Database connection pool established.

```
sara-abid@sara-abid-ThinkBook-14-G6-IRL: ~/smart_gym_management/backend
sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management/backend$ cd ~/smart_gym_management/backend
sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management/backend$ source venv/bin/activate
(venv) sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management/backend$ python test_auth.py
=====
SMART GYM API - AUTHENTICATION TESTS
=====
Make sure the Flask server is running on http://localhost:5000
Press Enter to start tests...

🔧 TEST 1: User Registration
=====
Register MEMBER
=====
Status Code: 201
Response:
{
  "message": "User registered successfully",
  "success": true,
  "user_id": 21
}
=====
Register TRAINER
=====
Status Code: 201
Response:
{
  "message": "User registered successfully",
  "success": true,
  "user_id": 22
}
=====
Register STAFF
=====
Status Code: 201
Response:
{
  "message": "User registered successfully",
  "success": true,
  "user_id": 23
}
=====
Register ADMIN
```

```
sara-abid@sara-abid-ThinkBook-14-G6-IRL: ~/smart_gym_management/backend
=====
Register ADMIN
=====
Status Code: 201
Response:
{
  "message": "User registered successfully",
  "success": true,
  "user_id": 24
}

🔧 TEST 2: Duplicate Email (Should Fail)
=====
Duplicate Registration
=====
Status Code: 409
Response:
{
  "error": "Email already registered",
  "success": false
}

🔧 TEST 3: Invalid Registration Data
--- Missing Password ---
=====
Missing Password
=====
Status Code: 400
Response:
{
  "error": "Missing required field: password",
  "success": false
}
--- Invalid Role ---
=====
Invalid Role
=====
Status Code: 400
Response:
{
  "error": "Invalid role. Must be one of: member, trainer, staff, admin",
  "success": false
}
```

```

sara-abid@sara-abid-ThinkBook-14-G6-IRL: ~/smart_gym_management/backend
sara-abid@sara-abid-ThinkBook-14-G6-IRL: ~/smart_gym_management/backend

--- Invalid Role ---
=====
Invalid Role
=====
Status Code: 400
Response:
{
  "error": "Invalid role. Must be one of: member, trainer, staff, admin",
  "success": false
}

--- Weak Password ---
=====
Weak Password
=====
Status Code: 400
Response:
{
  "error": "Password must be at least 8 characters long",
  "success": false
}

🔧 TEST 4: User Login (All Roles)
=====
Login: test.member@gym.com
=====
Status Code: 200
Response:
{
  "message": "Login successful",
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoyMSwiZWlhaWwiOiJ0ZXN0Ln1lbWJlc2BneW0uY29tIiwicm9sZSI6Im1lbWJlc2IsImV4cCI6MTc3Mjg4MDY4OCwiaWF0IjoxNzcyNzk0Mjg4fQ.DbEplgycKloCo8ZnQtYyW4AY130q1pu08nX1jh7G9d0",
  "user": {
    "email": "test.member@gym.com",
    "first_name": "Test",
    "last_name": "Member",
    "role": "member",
    "user_id": 21
  }
}

=====
Login: test.member@gym.com

```

```
sara-abid@sara-abid-ThinkBook-14-G6-IRL: ~/smart_gym_management/backend
=====
Login: test.trainer@gym.com
=====
Status Code: 200
Response:
{
  "message": "Login successful",
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoyMiwiaWwiOiJ0ZXN0LnR5YmVudGd5b55jb20iLCJyb2x1Ijoic3RhZmYlLCJleHAiOiJ0e3NzI4ODAwMjc5NDI5Mj00Lj8PyeR-RC30aYrUKTxr1leIiNXBPQRmou7lRkgxmEGD8",
  "user": {
    "email": "test.trainer@gym.com",
    "first_name": "Test",
    "last_name": "Trainer",
    "role": "trainer",
    "user_id": 22
  }
}
=====
Login: test.staff@gym.com
=====
Status Code: 200
Response:
{
  "message": "Login successful",
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoyMiwiaWwiOiJ0ZXN0LnR5YmVudGd5b55jb20iLCJyb2x1Ijoic3RhZmYlLCJleHAiOiJ0e3NzI4ODAwMjc5NDI5Mj00Lj8PyeR-kHVt6Y-vbaYnPNcknRVb9LqBtbl9luw",
  "user": {
    "email": "test.staff@gym.com",
    "first_name": "Test",
    "last_name": "Staff",
    "role": "staff",
    "user_id": 23
  }
}
=====
Login: test.admin@gym.com
=====
Status Code: 200
Response:
{
  "message": "Login successful",
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoyNCwiZW1haWwiOiJ0ZXN0LnR5YmVudGd5b55jb20iLCJyb2x1IjoieWVrtaW4iLCJleHAiOiJ0e3NzI4ODAwMjc5NDI5Mj00Lj8PyeR-FjovsdRetYxMZh1jrfw-AjGgf700a4oU",
  "user": {
    "email": "test.admin@gym.com",
    "first_name": "Test",
    "last_name": "Admin",
    "role": "admin",
    "user_id": 24
  }
}

🔧 TEST 5: Invalid Login Credentials
--- Wrong Password ---
=====
Wrong Password
=====
Status Code: 401
Response:
{
  "error": "Invalid email or password",
  "success": false
}
--- Non-existent Email ---
=====
Non-existent Email
=====
Status Code: 401
Response:
{
  "error": "Invalid email or password",
  "success": false
}
--- Missing Password ---
=====
Missing Password
=====
Status Code: 400
```

Authentication tested and verified.
Testing member endpoints:


```
(venv) sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management/backend$ python test_members.py
```

```
=====
SMART GYM API - MEMBER ENDPOINTS TESTS
=====
```

Make sure:

1. Flask server is running (python run.py)
2. Test users exist (run test_auth.py first)
3. Database has seed data

Press Enter to start tests...

🔑 Getting member authentication token...

✅ Token obtained: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkI...

🔧 TEST 1: Get Member Dashboard

=====
Member Dashboard
=====

Status Code: 200

Response:

```
{
  "dashboard": {
    "attended_classes": 0,
    "confirmed_bookings": 0,
    "email": "Test.member@gym.com",
    "full_name": "Test Member",
    "membership_expires": null,
    "membership_status": null,
    "membership_tier": null,
    "total_calories_burned": 0,
    "total_visits": 0,
    "total_workout_logs": 0,
    "user_id": 21
  },
  "success": true
}
```

🔧 TEST 2: Get Active Membership

=====
Active Membership
=====

Status Code: 404

Response:

🔧 TEST 2: Get Active Membership

=====
Active Membership
=====

Status Code: 404

Response:

```
{
  "error": "No active membership found",
  "message": "Please purchase a membership to access gym facilities",
  "success": false
}
```

🔧 TEST 3: Get Class Bookings

--- All Bookings ---

=====
All Bookings
=====

Status Code: 200

Response:

```
{
  "bookings": [],
  "count": 0,
  "success": true
}
```

--- Upcoming Bookings ---

=====
Upcoming Bookings
=====

Status Code: 200

Response:

```
{
  "bookings": [],
  "count": 0,
  "success": true
}
```

--- Confirmed Bookings ---

=====
Confirmed Bookings
=====

Status Code: 200

```
confirmed bookings
=====
Confirmed Bookings
=====
Status Code: 200
Response:
{
  "bookings": [],
  "count": 0,
  "success": true
}

🔗 TEST 4: Book a Class
=====
Book Class (schedule_id: 7)
=====
Status Code: 201
Response:
{
  "booking": {
    "booking_id": 32,
    "class_name": "Morning Yoga",
    "position_in_waitlist": null,
    "schedule_date": "2026-03-03",
    "start_time": "06:00:00",
    "status": "confirmed"
  },
  "message": "Class booked successfully",
  "success": true
}

--- Try Duplicate Booking ---
=====
Duplicate Booking (Should Fail)
=====
Status Code: 409
Response:
{
  "error": "You have already booked this class",
  "existing_status": "confirmed",
  "success": false
}

🔗 TEST 5: Cancel a Booking
```

```
✓ TEST 5: Cancel a Booking
No bookings found to cancel. Book a class first.
```

✓ TEST 6: Get Workout Logs

```
--- All Workouts ---
```

```
=====
All Workouts
```

```
=====
Status Code: 200
```

```
Response:
```

```
{
  "count": 0,
  "success": true,
  "workouts": []
}
```

```
--- Workouts (Last 7 days) ---
```

```
=====
Recent Workouts
```

```
=====
Status Code: 200
```

```
Response:
```

```
{
  "count": 0,
  "success": true,
  "workouts": []
}
```

✓ TEST 7: Log a Workout

```
=====
Log Workout
```

```
=====
Status Code: 201
```

```
Response:
```

```
{
  "log_id": 40,
  "message": "Workout logged successfully",
  "success": true
}
```

```
--- Minimal Workout Log ---
```

```
--- Minimal Workout Log ---
```

```
=====
Minimal Workout Log
```

```
=====
Status Code: 201
```

```
Response:
```

```
{
  "log_id": 41,
  "message": "Workout logged successfully",
  "success": true
}
```

✓ TEST 8: Get Profile

```
=====
Get Profile
```

```
=====
Status Code: 200
```

```
Response:
```

```
{
  "profile": {
    "created_at": "2026-03-06T15:51:26.838590",
    "date_of_birth": "1995-05-15",
    "email": "test.member@gym.com",
    "first_name": "Test",
    "gender": "male",
    "last_name": "Member",
    "phone": "0300-9999999",
    "user_id": 21
  },
  "success": true
}
```

✓ TEST 9: Update Profile

```
=====
Update Profile
```

```
=====
Status Code: 200
```

```
Response:
```

```
{
  "message": "Profile updated successfully",
  "success": true
}
```

```
--- Try Update Invalid Field ---
```

```
=====
Invalid Field Update (Should Be Ignored)
=====
Status Code: 400
Response:
{
  "allowed_fields": [
    "first_name",
    "last_name",
    "phone",
    "date_of_birth",
    "gender"
  ],
  "error": "No valid fields to update",
  "success": false
}
```

✔ TEST 10: Unauthorized Access

--- No Token ---

```
=====
Access Without Token (Should Fail)
=====
Status Code: 401
Response:
{
  "error": "Authorization token is missing",
  "success": false
}
```

--- Invalid Token ---

```
=====
Access With Invalid Token (Should Fail)
=====
Status Code: 401
Response:
{
  "error": "Invalid token. Please login again.",
  "success": false
}
```

✔ TEST 11: Wrong Role Access

✔ TEST 10: Unauthorized Access

--- No Token ---

```
=====
Access Without Token (Should Fail)
=====
Status Code: 401
Response:
{
  "error": "Authorization token is missing",
  "success": false
}
```

--- Invalid Token ---

```
=====
Access With Invalid Token (Should Fail)
=====
Status Code: 401
Response:
{
  "error": "Invalid token. Please login again.",
  "success": false
}
```

✔ TEST 11: Wrong Role Access

```
=====
Trainer Accessing Member Endpoint (Should Fail)
=====
Status Code: 403
Response:
{
  "error": "Access denied. Required role(s): member",
  "success": false,
  "your_role": "trainer"
}
```

✔ ALL MEMBER ENDPOINT TESTS COMPLETED

Review the results above.
Expected: Most tests should return 200/201
Expected failures: unauthorized access, wrong role
(env) [sara-ahid@sara-ahid-ThinkBook-14-65-T81 > smart_num_management/backends](#)

Testing admin role and general

```
sara-abid@sara-abid-ThinkBook-14-G6-IRL: /smart_gyn_management$ ./test_admin_api.sh
---- Get all users ----
{
  "pagination": {
    "page": 1,
    "per_page": 10,
    "total": 25,
    "total_pages": 3
  },
  "success": true,
  "users": [
    {
      "created_at": "2026-03-08T05:00:33.629714",
      "date_of_birth": null,
      "email": "admin@gym.com",
      "first_name": "Admin",
      "gender": null,
      "is_active": true,
      "last_name": "User",
      "phone": null,
      "role": "admin",
      "user_id": 25
    },
    {
      "created_at": "2026-03-06T15:51:28.101237",
      "date_of_birth": null,
      "email": "test.admin@gym.com",
      "first_name": "Test",
      "gender": null,
      "is_active": true,
      "last_name": "Admin",
      "phone": null,
      "role": "admin",
      "user_id": 24
    },
    {
      "created_at": "2026-03-06T15:51:27.698935",
      "date_of_birth": null,
      "email": "test.staff@gym.com",
      "first_name": "Test",
      "gender": "male",
      "is_active": true,
      "last_name": "Staff",
      "phone": "0302-9999999",
      "role": "staff",
      "user_id": 23
    }
  ]
}
```

```

{
  "created_at": "2026-03-06T15:51:27.698935",
  "date_of_birth": null,
  "email": "test.staff@gym.com",
  "first_name": "Test",
  "gender": "male",
  "is_active": true,
  "last_name": "Staff",
  "phone": "0302-9999999",
  "role": "staff",
  "user_id": 23
},
{
  "created_at": "2026-03-06T15:51:27.299703",
  "date_of_birth": "1988-08-20",
  "email": "test.trainer@gym.com",
  "first_name": "Test",
  "gender": "female",
  "is_active": true,
  "last_name": "Trainer",
  "phone": "0301-9999999",
  "role": "trainer",
  "user_id": 22
},
{
  "created_at": "2026-03-06T15:51:26.838590",
  "date_of_birth": "1995-05-15",
  "email": "test.member@gym.com",
  "first_name": "Ahmed",
  "gender": "male",
  "is_active": true,
  "last_name": "Khan",
  "phone": "0300-9999999",
  "role": "member",
  "user_id": 21
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1994-12-05",
  "email": "hamza.raza@gmail.com",
  "first_name": "Hamza",
  "gender": "male",
  "is_active": true,
  "last_name": "Raza",
  "phone": "0306-7890123",
  "role": "member",
  "user_id": 7
},
{
  "created_at": "2026-02-22T01:05:59.285380",

```

```

},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1997-01-30",
  "email": "zainab.hassan@gmail.com",
  "first_name": "Zainab",
  "gender": "female",
  "is_active": true,
  "last_name": "Hassan",
  "phone": "0305-6789012",
  "role": "member",
  "user_id": 6
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1992-11-10",
  "email": "bilal.ahmed@gmail.com",
  "first_name": "Bilal",
  "gender": "male",
  "is_active": true,
  "last_name": "Ahmed",
  "phone": "0302-3456789",
  "role": "member",
  "user_id": 3
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1996-02-28",
  "email": "maria.farooq@gmail.com",
  "first_name": "Maria",
  "gender": "female",
  "is_active": true,
  "last_name": "Farooq",
  "phone": "0309-0123456",
  "role": "member",
  "user_id": 10
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1990-09-25",
  "email": "usman.tariq@gmail.com",
  "first_name": "Usman",
  "gender": "male",
  "is_active": true,
  "last_name": "Tariq",
  "phone": "0304-5678901",
  "role": "member",
  "user_id": 5
}
}

```

```
    }
  }
}

---- Search users ----
{
  "pagination": {
    "page": 1,
    "per_page": 20,
    "total": 3,
    "total_pages": 1
  },
  "success": true,
  "users": [
    {
      "created_at": "2026-03-06T15:51:26.838590",
      "date_of_birth": "1995-05-15",
      "email": "test.member@gym.com",
      "first_name": "Ahmed",
      "gender": "male",
      "is_active": true,
      "last_name": "Khan",
      "phone": "0300-9999999",
      "role": "member",
      "user_id": 21
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1995-03-15",
      "email": "ahmed.khan@gmail.com",
      "first_name": "Ahmed",
      "gender": "male",
      "is_active": true,
      "last_name": "Khan",
      "phone": "0300-1234567",
      "role": "member",
      "user_id": 1
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1992-11-10",
      "email": "bilal.ahmed@gmail.com",
      "first_name": "Bilal",
      "gender": "male",
      "is_active": true,
      "last_name": "Ahmed",
      "phone": "0302-3456789",
      "role": "member"
    }
  ]
}
```



```

    "date_of_birth": "1992-11-10",
    "email": "bilal.ahmed@gmail.com",
    "first_name": "Bilal",
    "gender": "male",
    "is_active": true,
    "last_name": "Ahmed",
    "phone": "0302-3456789",
    "role": "member",
    "user_id": 3
  }
]
}

---- Filter by role ----
{
  "pagination": {
    "page": 1,
    "per_page": 20,
    "total": 13,
    "total_pages": 1
  },
  "success": true,
  "users": [
    {
      "created_at": "2026-03-06T15:51:26.838590",
      "date_of_birth": "1995-05-15",
      "email": "test.member@gym.com",
      "first_name": "Ahmed",
      "gender": "male",
      "is_active": true,
      "last_name": "Khan",
      "phone": "0300-9999999",
      "role": "member",
      "user_id": 21
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1998-07-22",
      "email": "fatima.ali@gmail.com",
      "first_name": "Fatima",
      "gender": "female",
      "is_active": true,
      "last_name": "Ali",
      "phone": "0301-2345678",
      "role": "member",
      "user_id": 2
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1992-11-10",
      "email": "bilal.ahmed@gmail.com",
      "first_name": "Bilal",
      "gender": "male",
      "is_active": true,
      "last_name": "Ahmed",
      "phone": "0302-3456789",
      "role": "member",
      "user_id": 3
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "2000-05-18",
      "email": "ayesha.malik@gmail.com",
      "first_name": "Ayesha",
      "gender": "female",
      "is_active": true,
      "last_name": "Malik",
      "phone": "0303-4567890",
      "role": "member",
      "user_id": 4
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1990-09-25",
      "email": "usman.tariq@gmail.com",
      "first_name": "Usman",
      "gender": "male",
      "is_active": true,
      "last_name": "Tariq",
      "phone": "0304-5678901",
      "role": "member",
      "user_id": 5
    }
  ]
}

```

```

    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1998-07-22",
      "email": "fatima.ali@gmail.com",
      "first_name": "Fatima",
      "gender": "female",
      "is_active": true,
      "last_name": "Ali",
      "phone": "0301-2345678",
      "role": "member",
      "user_id": 2
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1992-11-10",
      "email": "bilal.ahmed@gmail.com",
      "first_name": "Bilal",
      "gender": "male",
      "is_active": true,
      "last_name": "Ahmed",
      "phone": "0302-3456789",
      "role": "member",
      "user_id": 3
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "2000-05-18",
      "email": "ayesha.malik@gmail.com",
      "first_name": "Ayesha",
      "gender": "female",
      "is_active": true,
      "last_name": "Malik",
      "phone": "0303-4567890",
      "role": "member",
      "user_id": 4
    },
    {
      "created_at": "2026-02-22T01:05:59.285380",
      "date_of_birth": "1990-09-25",
      "email": "usman.tariq@gmail.com",
      "first_name": "Usman",
      "gender": "male",
      "is_active": true,
      "last_name": "Tariq",
      "phone": "0304-5678901",
      "role": "member",
      "user_id": 5
    }
  ]
}

```

```

    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1990-09-25",
    "email": "usman.tariq@gmail.com",
    "first_name": "Usman",
    "gender": "male",
    "is_active": true,
    "last_name": "Tariq",
    "phone": "0304-5678901",
    "role": "member",
    "user_id": 5
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1997-01-30",
    "email": "zainab.hassan@gmail.com",
    "first_name": "Zainab",
    "gender": "female",
    "is_active": true,
    "last_name": "Hassan",
    "phone": "0305-6789012",
    "role": "member",
    "user_id": 6
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1995-03-15",
    "email": "ahmed.khan@gmail.com",
    "first_name": "Ahmed",
    "gender": "male",
    "is_active": true,
    "last_name": "Khan",
    "phone": "0300-1234567",
    "role": "member",
    "user_id": 1
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1999-04-12",
    "email": "sana.shah@gmail.com",
    "first_name": "Sana",
    "gender": "female",
    "is_active": true,
    "last_name": "Shah",
    "phone": "0307-8901234",
    "role": "member",
    "user_id": 8
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",

```

```

    "user_id": 8
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1991-08-20",
    "email": "imran.queshi@gmail.com",
    "first_name": "Imran",
    "gender": "male",
    "is_active": true,
    "last_name": "Qureshi",
    "phone": "0308-9012345",
    "role": "member",
    "user_id": 9
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1996-02-28",
    "email": "maria.farooq@gmail.com",
    "first_name": "Maria",
    "gender": "female",
    "is_active": true,
    "last_name": "Farooq",
    "phone": "0309-0123456",
    "role": "member",
    "user_id": 10
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "1993-06-14",
    "email": "ali.rauf@gmail.com",
    "first_name": "Ali",
    "gender": "male",
    "is_active": true,
    "last_name": "Rauf",
    "phone": "0310-1234567",
    "role": "member",
    "user_id": 11
  },
  {
    "created_at": "2026-02-22T01:05:59.285380",
    "date_of_birth": "2001-10-08",
    "email": "hira.iqbal@gmail.com",
    "first_name": "Hira",
    "gender": "female",
    "is_active": true,
    "last_name": "Iqbal",
    "phone": "0311-2345678",
    "role": "member",
    "user_id": 12
  },

```

```
-
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1993-06-14",
  "email": "ali.rauf@gmail.com",
  "first_name": "Ali",
  "gender": "male",
  "is_active": true,
  "last_name": "Rauf",
  "phone": "0310-1234567",
  "role": "member",
  "user_id": 11
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "2001-10-08",
  "email": "hira.iqbal@gmail.com",
  "first_name": "Hira",
  "gender": "female",
  "is_active": true,
  "last_name": "Iqbal",
  "phone": "0311-2345678",
  "role": "member",
  "user_id": 12
},
{
  "created_at": "2026-02-22T01:05:59.285380",
  "date_of_birth": "1994-12-05",
  "email": "hanza.raza@gmail.com",
  "first_name": "Hanza",
  "gender": "male",
  "is_active": true,
  "last_name": "Raza",
  "phone": "0306-7890123",
  "role": "member",
  "user_id": 7
}
]
}

---- Create new user ----
{
  "message": "User created successfully",
  "success": true,
  "user_id": 26
}

---- Update user ----
{
```

```

---- Create new user ----
{
  "message": "User created successfully",
  "success": true,
  "user_id": 26
}

---- Update user ----
{
  "message": "User updated successfully",
  "success": true
}

---- Deactivate user ----
{
  "message": "User deactivated successfully",
  "success": true
}

---- Revenue report ----
{
  "revenue": {
    "by_payment_method": {
      "cash": 9000.0,
      "credit_card": 30000.0,
      "debit_card": 7500.0,
      "online": 16000.0
    },
    "late_fee_revenue": 800.0,
    "membership_revenue": 46000.0,
    "period": {
      "end_date": "2026-02-21",
      "start_date": "2026-01-01"
    },
    "product_revenue": 7800.0,
    "total_revenue": 63100.0,
    "training_revenue": 8500.0
  },
  "success": true
}

---- System analytics ----
{
  "analytics": {

```

```

---- System analytics ----
{
  "analytics": {
    "class_stats": {
      "avg_capacity": 5.4,
      "completed": 3,
      "total_schedules": 15,
      "upcoming": 12
    },
    "equipment_stats": {
      "maintenance_due_soon": 3,
      "maintenance_overdue": 3,
      "total": 15
    },
    "membership_stats": {
      "active": 12,
      "expired": 4,
      "frozen": 2
    },
    "user_stats": {
      "admin": {
        "active": 3,
        "total": 3
      },
      "member": {
        "active": 13,
        "total": 14
      },
      "staff": {
        "active": 3,
        "total": 3
      },
      "trainer": {
        "active": 6,
        "total": 6
      }
    }
  },
  "success": true
}

---- Memberships ----
{
  "memberships": [
    {
      "auto_renew": true,
      "email": "sana.shah@gmail.com",
      "membership_id": 1,
      "status": "active"
    }
  ]
}

```

```
---- Memberships ----
{
  "memberships": [
    {
      "auto_renew": true,
      "email": "sana.shah@gmail.com",
      "end_date": "2027-02-10",
      "is_frozen": false,
      "member_name": "Sana Shah",
      "membership_id": 8,
      "start_date": "2026-02-10",
      "status": "active",
      "tier_name": "Basic",
      "user_id": 8
    },
    {
      "auto_renew": true,
      "email": "ali.rauf@gmail.com",
      "end_date": "2027-02-05",
      "is_frozen": false,
      "member_name": "Ali Rauf",
      "membership_id": 11,
      "start_date": "2026-02-05",
      "status": "active",
      "tier_name": "Basic",
      "user_id": 11
    },
    {
      "auto_renew": true,
      "email": "usman.tariq@gmail.com",
      "end_date": "2027-01-31",
      "is_frozen": false,
      "member_name": "Updated Tariq",
      "membership_id": 5,
      "start_date": "2026-02-01",
      "status": "active",
      "tier_name": "Basic",
      "user_id": 5
    },
    {
      "auto_renew": true,
      "email": "fatima.ali@gmail.com",
      "end_date": "2027-01-31",
      "is_frozen": false,
      "member_name": "Fatima Ali",
      "membership_id": 2,
      "start_date": "2026-02-01",
      "status": "active",
      "tier_name": "VIP"
    }
  ]
}
```

```
{
  "auto_renew": true,
  "email": "fatima.ali@gmail.com",
  "end_date": "2027-01-31",
  "is_frozen": false,
  "member_name": "Fatima Ali",
  "membership_id": 2,
  "start_date": "2026-02-01",
  "status": "active",
  "tier_name": "VIP",
  "user_id": 2
},
{
  "auto_renew": true,
  "email": "hira. iqbal@gmail.com",
  "end_date": "2027-01-25",
  "is_frozen": false,
  "member_name": "Hira Iqbal",
  "membership_id": 12,
  "start_date": "2026-01-25",
  "status": "active",
  "tier_name": "Premium",
  "user_id": 12
},
{
  "auto_renew": true,
  "email": "maria.farooq@gmail.com",
  "end_date": "2027-01-20",
  "is_frozen": false,
  "member_name": "Maria Farooq",
  "membership_id": 10,
  "start_date": "2026-01-20",
  "status": "active",
  "tier_name": "VIP",
  "user_id": 10
},
{
  "auto_renew": true,
  "email": "hamza.raza@gmail.com",
  "end_date": "2027-01-15",
  "is_frozen": false,
  "member_name": "Hamza Raza",
  "membership_id": 7,
  "start_date": "2026-01-15",
  "status": "active",
  "tier_name": "Premium",
  "user_id": 7
},
}
```

```

    "auto_renew": true,
    "email": "hamza.raza@gmail.com",
    "end_date": "2027-01-15",
    "is_frozen": false,
    "member_name": "Hamza Raza",
    "membership_id": 7,
    "start_date": "2026-01-15",
    "status": "active",
    "tier_name": "Premium",
    "user_id": 7
  },
  {
    "auto_renew": true,
    "email": "ayesha.malik@gmail.com",
    "end_date": "2027-01-10",
    "is_frozen": false,
    "member_name": "Ayesha Malik",
    "membership_id": 4,
    "start_date": "2026-01-10",
    "status": "active",
    "tier_name": "Premium",
    "user_id": 4
  },
  {
    "auto_renew": true,
    "email": "ahmed.khan@gmail.com",
    "end_date": "2026-12-31",
    "is_frozen": false,
    "member_name": "Ahmed Khan",
    "membership_id": 1,
    "start_date": "2026-01-01",
    "status": "active",
    "tier_name": "Premium",
    "user_id": 1
  },
  {
    "auto_renew": true,
    "email": "bilal.ahmed@gmail.com",
    "end_date": "2026-11-15",
    "is_frozen": false,
    "member_name": "Bilal Ahmed",
    "membership_id": 3,
    "start_date": "2025-12-15",
    "status": "active",
    "tier_name": "Basic",
    "user_id": 3
  },
  {
    "auto_renew": true,

```

```

    "auto_renew": true,
    "email": "imran.queshi@gmail.com",
    "end_date": "2026-11-30",
    "is_frozen": false,
    "member_name": "Imran Qureshi",
    "membership_id": 9,
    "start_date": "2025-12-01",
    "status": "active",
    "tier_name": "Premium",
    "user_id": 9
  },
  {
    "auto_renew": true,
    "email": "zainab.hassan@gmail.com",
    "end_date": "2026-10-31",
    "is_frozen": false,
    "member_name": "Zainab Hassan",
    "membership_id": 6,
    "start_date": "2025-11-01",
    "status": "active",
    "tier_name": "VIP",
    "user_id": 6
  }
],
"pagination": {
  "page": 1,
  "per_page": 20,
  "total": 12,
  "total_pages": 1
},
"success": true
}

```

---- Equipment health ----

```

{
  "count": 15,
  "equipment": [
    {
      "equipment_id": 12,
      "equipment_name": "Dumbbell Set",
      "equipment_type": "Strength",
      "last_maintenance_date": "2025-12-20",
      "maintenance_alert": "OVERDUE",
      "next_maintenance_due": "2026-02-20",
      "status": "available",
      "total_maintenance_cost": 0,
      "total_maintenance_records": 0,
      "total_usage_hours": 850,
    }
  ]
}

```

```
---- Equipment Health ----
{
  "count": 15,
  "equipment": [
    {
      "equipment_id": 12,
      "equipment_name": "Dumbbell Set",
      "equipment_type": "Strength",
      "last_maintenance_date": "2025-12-20",
      "maintenance_alert": "OVERDUE",
      "next_maintenance_due": "2026-02-20",
      "status": "available",
      "total_maintenance_cost": 0,
      "total_maintenance_records": 0,
      "total_usage_hours": 850,
      "usage_category": "HIGH USAGE",
      "zone": "Free Weights Zone"
    },
    {
      "equipment_id": 7,
      "equipment_name": "Squat Rack 01",
      "equipment_type": "Strength",
      "last_maintenance_date": "2026-01-05",
      "maintenance_alert": "OVERDUE",
      "next_maintenance_due": "2026-03-05",
      "status": "available",
      "total_maintenance_cost": 4000.0,
      "total_maintenance_records": 1,
      "total_usage_hours": 680,
      "usage_category": "HIGH USAGE",
      "zone": "Strength Zone"
    },
    {
      "equipment_id": 8,
      "equipment_name": "Squat Rack 02",
      "equipment_type": "Strength",
      "last_maintenance_date": "2026-01-05",
      "maintenance_alert": "OVERDUE",
      "next_maintenance_due": "2026-03-05",
      "status": "available",
      "total_maintenance_cost": 4000.0,
      "total_maintenance_records": 1,
      "total_usage_hours": 620,
      "usage_category": "HIGH USAGE",
      "zone": "Strength Zone"
    },
    {
      "equipment_id": 9,
      "equipment_name": "Bench Press 01",
      "equipment_type": "Strength",
      "last_maintenance_date": "2025-12-20",
      "maintenance_alert": "OVERDUE",
      "next_maintenance_due": "2026-02-20",
      "status": "available",
      "total_maintenance_cost": 0,
      "total_maintenance_records": 0,
      "total_usage_hours": 850,
      "usage_category": "HIGH USAGE",
      "zone": "Free Weights Zone"
    }
  ]
}
```



```

    "equipment_id": 1,
    "equipment_name": "Treadmill 01",
    "equipment_type": "Cardio",
    "last_maintenance_date": "2026-01-10",
    "maintenance_alert": "DUE SOON",
    "next_maintenance_due": "2026-03-10",
    "status": "available",
    "total_maintenance_cost": 5000.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 450,
    "usage_category": "MODERATE USAGE",
    "zone": "Cardio Zone"
  },
  {
    "equipment_id": 2,
    "equipment_name": "Treadmill 02",
    "equipment_type": "Cardio",
    "last_maintenance_date": "2026-01-10",
    "maintenance_alert": "DUE SOON",
    "next_maintenance_due": "2026-03-10",
    "status": "available",
    "total_maintenance_cost": 5000.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 380,
    "usage_category": "MODERATE USAGE",
    "zone": "Cardio Zone"
  },
  {
    "equipment_id": 14,
    "equipment_name": "Battle Ropes",
    "equipment_type": "Functional",
    "last_maintenance_date": "2026-02-15",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-15",
    "status": "available",
    "total_maintenance_cost": 0,
    "total_maintenance_records": 0,
    "total_usage_hours": 95,
    "usage_category": "LOW USAGE",
    "zone": "Functional Training Zone"
  },
  {
    "equipment_id": 10,
    "equipment_name": "Cable Machine 01",
    "equipment_type": "Strength",
    "last_maintenance_date": "2026-02-05",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-05",
    "status": "available"
  }

```

```

    "equipment_name": "Cable Machine 01",
    "equipment_type": "Strength",
    "last_maintenance_date": "2026-02-05",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-05",
    "status": "available",
    "total_maintenance_cost": 12000.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 410,
    "usage_category": "MODERATE USAGE",
    "zone": "Strength Zone"
  },
  {
    "equipment_id": 5,
    "equipment_name": "Elliptical 01",
    "equipment_type": "Cardio",
    "last_maintenance_date": "2026-01-25",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-03-25",
    "status": "available",
    "total_maintenance_cost": 2000.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 290,
    "usage_category": "MODERATE USAGE",
    "zone": "Cardio Zone"
  },
  {
    "equipment_id": 13,
    "equipment_name": "Kettlebell Set",
    "equipment_type": "Functional",
    "last_maintenance_date": "2026-02-01",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-01",
    "status": "available",
    "total_maintenance_cost": 0,
    "total_maintenance_records": 0,
    "total_usage_hours": 280,
    "usage_category": "MODERATE USAGE",
    "zone": "Functional Training Zone"
  },
  {
    "equipment_id": 11,
    "equipment_name": "Leg Press 01",
    "equipment_type": "Strength",
    "last_maintenance_date": "2026-02-10",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-10",
    "status": "available",
    "total_maintenance_cost": 0,
  }

```

```
{
    "total_maintenance_cost": 0,
    "total_maintenance_records": 0,
    "total_usage_hours": 360,
    "usage_category": "MODERATE USAGE",
    "zone": "Strength Zone"
},
{
    "equipment_id": 15,
    "equipment_name": "Pull-up Bar Station",
    "equipment_type": "Functional",
    "last_maintenance_date": "2026-01-20",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-03-20",
    "status": "available",
    "total_maintenance_cost": 0,
    "total_maintenance_records": 0,
    "total_usage_hours": 430,
    "usage_category": "MODERATE USAGE",
    "zone": "Functional Training Zone"
},
{
    "equipment_id": 6,
    "equipment_name": "Rowing Machine 01",
    "equipment_type": "Cardio",
    "last_maintenance_date": "2026-02-01",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-04-01",
    "status": "available",
    "total_maintenance_cost": 2500.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 180,
    "usage_category": "LOW USAGE",
    "zone": "Cardio Zone"
},
{
    "equipment_id": 4,
    "equipment_name": "Stationary Bike 01",
    "equipment_type": "Cardio",
    "last_maintenance_date": "2026-01-20",
    "maintenance_alert": "OK",
    "next_maintenance_due": "2026-03-20",
    "status": "available",
    "total_maintenance_cost": 3000.0,
    "total_maintenance_records": 1,
    "total_usage_hours": 310,
    "usage_category": "MODERATE USAGE",
    "zone": "Cardio Zone"
},
}
```

```
📄 Health Check: http://localhost:5000/health
🔑 Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.100.11:5000
Press CTRL+C to quit
* Restarting with stat
🟢 Database connection pool created successfully

🚀 Starting Smart Gym API on http://localhost:5000
📄 API Documentation: http://localhost:5000/api/docs (coming soon)
📄 Health Check: http://localhost:5000/health
🔑 Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Debugger is active!
* Debugger PIN: 121-210-517
127.0.0.1 - - [08/Mar/2026 05:00:33] "POST /api/v1/auth/register HTTP/1.1" 201 -
127.0.0.1 - - [08/Mar/2026 05:00:34] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:02:38] "GET /api/v1/admin/users?page=1 HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:02:38] "GET /api/v1/admin/users?search=ahmed HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:02:38] "GET /api/v1/admin/users?role=member HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:02:38] "POST /api/v1/admin/users HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:02:39] "PUT /api/v1/admin/users/5 HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:02:39] "DELETE /api/v1/admin/users/5 HTTP/1.1" 401 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/users?page=1&per_page=10 HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/users?search=ahmed HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/users?role=member HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "POST /api/v1/admin/users HTTP/1.1" 201 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "PUT /api/v1/admin/users/5 HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "DELETE /api/v1/admin/users/5 HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/revenue?start_date=2026-01-01&end_date=2026-02-21 HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/analytics HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/memberships?status=active HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/admin/equipment-health HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/classes HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/classes/schedule HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/classes/schedule?date=2026-02-24 HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/membership-tiers HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/trainers HTTP/1.1" 200 -
127.0.0.1 - - [08/Mar/2026 05:14:24] "GET /api/v1/trainers?available=true HTTP/1.1" 200 -
```

```
(venv) ser_123-holla@LAPTOP-60Q3PWS4:/mnt/c/Users/User/Desktop/smart_gym_and_fitness_project/Smart-Gym-Fitness-Center-Management-System/backend$ python run.py
Database connection pool created successfully

Starting Smart Gym API on http://localhost:5000
API Documentation: http://localhost:5000/api/docs (coming soon)
Health Check: http://localhost:5000/health
Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.24.102.194:5000
Press CTRL+C to quit
* Restarting with stat
Database connection pool created successfully

Starting Smart Gym API on http://localhost:5000
API Documentation: http://localhost:5000/api/docs (coming soon)
Health Check: http://localhost:5000/health
Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Debugger is active!
* Debugger PIN: 111-100-485
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 404 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 401 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/payments/history HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/schedule HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/ HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/schedule HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 401 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 403 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 404 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "DELETE /api/v1/classes/cancel/54 HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "DELETE /api/v1/classes/cancel/54 HTTP/1.1" 400 -
]
```

```
• (venv) ser_123-holla@LAPTOP-6DQ3PMS4:/mnt/c/Users/User/Desktop/smart_gym_and_fitness_project/Smart-Gym-Fitness-Center-
```

SMART GYM – STEP 4 TRANSACTION TESTS

Make sure you ran test_auth.py first to create test.member@gym.com
Make sure Flask is running (python run.py). Press Enter...

Auto-cleaning test data from previous runs...

Removed 2 test booking(s) for ahmed

Removed 1 test membership(s) for test.member

Removed 1 test payment(s) for test.member

Cleanup done – starting fresh!

```
=====
  SETUP: Getting tokens
=====
```

Member 1 (ahmed.khan) logged in

Member 2 (test.member) logged in

```
=====
  SCENARIO 1: MEMBERSHIP PURCHASE TRANSACTION
=====
```

Testing: Atomicity, Consistency, SELECT FOR UPDATE (locker), Durability

T01 | Missing tier_id → 400

Expected: 400 | Got: 400

{

"error": "tier_id is required",

"success": false

}

T02 | Invalid payment_method → 400

Expected: 400 | Got: 400

{

"error": "payment_method must be one of: ['cash', 'credit_card', 'debit_card', 'online']",

"success": false

}

T03 | Invalid tier_id=999 → 404 (transaction rolled back)

Expected: 404 | Got: 404

{

"error": "Membership tier 999 does not exist",

"success": false

}

T04 | No token → 401

Expected: 401 | Got: 401

{

"error": "Authorization token is missing",

"success": false

}

T05 | Member 1 already has membership → 409

Expected: 409 | Got: 409

{

"error": "You already have an active membership",

"existing_membership_id": 1,

"success": false

}

(venv) ser_123-h011a@LAPTOP-6DQ3PMS4:/mnt/C:/Users/User/Desktop/smart_gym_and_fitness_project/SmartGym

T06 | Member 2 (test.member) purchases Basic membership...

T06 | Purchase Basic membership → 201

Expected: 201 | Got: 201

```
{
  "message": "Membership purchased successfully",
  "success": true,
  "transaction": {
    "locker": null,
    "membership": {
      "end_date": "2026-04-11",
      "membership_id": 25,
      "start_date": "2026-03-11",
      "status": "active",
      "tier": "Basic"
    },
    "payment": {
      "amount": 5000.0,
      "method": "credit_card",
      "payment_date": "2026-03-11 01:18:32.483095",
      "payment_id": 27,
      "status": "completed"
    }
  }
}
```

Payment ID : 27

Membership : 25 (Basic)

Valid until : 2026-04-11

Locker : None (None for Basic – correct)

T07 | Duplicate membership → 409

Expected: 409 | Got: 409

```
{
  "error": "You already have an active membership",
  "existing_membership_id": 25,
  "success": false
}
```

T08 | Payment history → 200

Expected: 200 | Got: 200

```
{
  "count": 2,
  "payments": [
    {
      "amount": "2000.00",
      "description": "1-hour PT with Coach Ali",
      "payment_date": "Tue, 10 Feb 2026 16:00:00 GMT",
      "payment_id": 9,
      "payment_method": "credit_card",
      "payment_type": "training_session",
      "status": "completed"
    },
    {
      "amount": "8000.00",
      "description": "Premium membership - Jan 2026",
      "payment_date": "Thu, 01 Jan 2026 10:00:00 GMT",
      "payment_id": 1,
      "payment_method": "credit_card",
      "payment_type": "membership",
      "status": "completed"
    }
  ]
}
```

```
    ],  
    "success": true  
}
```

SCENARIO 2: CLASS BOOKING TRANSACTION

Testing: SELECT FOR UPDATE, trigger integration, waitlist logic

Fetching class schedules...

Found 4 upcoming schedules

Open: 3 | Full: 1

→ ID:16 Morning Yoga 2026-03-13 (15/15 spots)

→ ID:17 HIIT Blast 2026-03-26 (20/20 spots)

→ ID:18 Zumba Dance 2026-04-12 (25/25 spots)

T09 | Get classes list → 200

Expected: 200 | Got: 200

```
{  
  "classes": [  
    {  
      "class_id": 6,  
      "class_name": "Beginner Strength",  
      "class_type": "Strength",  
      "description": "Introduction to weight training",  
      "difficulty_level": "beginner",  
      "duration_minutes": 60,  
      "max_capacity": 10  
    },  
    {  
      "class_id": 4,  
      "class_name": "CrossFit WOD",  
      "class_type": "Strength",  
      "description": "Workout of the Day - functional fitness",  
      "difficulty_level": "advanced",  
      "duration_minutes": 60,  
      "max_capacity": 12  
    },  
    {  
      "class_id": 2,  
      "class_name": "HIIT Blast",  
      "class_type": "Cardio",  
      "description": "High-intensity interval training for fat loss",  
      "difficulty_level": "advanced",  
      "duration_minutes": 45,  
      "max_capacity": 20  
    },  
    {  
      "class_id": 1,  
      "class_name": "Morning Yoga",  
      "class_type": "Mind & Body",  
      "description": "Relaxing yoga session to start your day",  
      "difficulty_level": "beginner",  
      "duration_minutes": 60,  
      "max_capacity": 15  
    },  
    {  
      "class_id": 5,  
      "class_name": "Spin Cycle",  
      "class_type": "Cardio",  
      "description": "Indoor cycling with motivating music",  
      "difficulty_level": "beginner",  
      "duration_minutes": 45,  
      "max_capacity": 25  
    }  
  ]  
}
```

```

    },
    {
      "class_id": 5,
      "class_name": "Spin Cycle",
      "class_type": "Cardio",
      "description": "Indoor cycling with motivating music",
      "difficulty_level": "intermediate",
      "duration_minutes": 45,
      "max_capacity": 18
    },
    {
      "class_id": 3,
      "class_name": "Zumba Dance",
      "class_type": "Cardio",
      "description": "Fun Latin-inspired dance workout",
      "difficulty_level": "intermediate",
      "duration_minutes": 50,
      "max_capacity": 25
    }
  ],
  "count": 6,
  "success": true
}

```

T10 | Get schedule → 200

Expected: 200 | Got: 200

```

{
  "count": 4,
  "schedules": [
    {
      "class_name": "Morning Yoga",
      "class_type": "Mind & Body",
      "current_capacity": 0,
      "difficulty_level": "beginner",
      "duration_minutes": 60,
      "end_time": "07:00:00",
      "max_capacity": 15,
      "room_number": "Studio A",
      "schedule_date": "2026-03-13",
      "schedule_id": 16,
      "spots_available": 15,
      "start_time": "06:00:00",
      "status": "scheduled",
      "trainer_name": "Sara Nadeem"
    },
    {
      "class_name": "HIIT Blast",
      "class_type": "Cardio",
      "current_capacity": 0,
      "difficulty_level": "advanced",
      "duration_minutes": 45,
      "end_time": "18:45:00",
      "max_capacity": 20,
      "room_number": "Studio B",
      "schedule_date": "2026-03-26",
      "schedule_id": 17,
      "spots_available": 20,
      "start_time": "18:00:00",
      "status": "scheduled",
      "trainer_name": "Nida Khan"
    }
  ],
}

```



```

    },
    {
      "class_name": "Zumba Dance",
      "class_type": "Cardio",
      "current_capacity": 0,
      "difficulty_level": "intermediate",
      "duration_minutes": 50,
      "end_time": "19:50:00",
      "max_capacity": 25,
      "room_number": "Studio A",
      "schedule_date": "2026-04-12",
      "schedule_id": 18,
      "spots_available": 25,
      "start_time": "19:00:00",
      "status": "scheduled",
      "trainer_name": "Sara Nadeem"
    },
    {
      "class_name": "Beginner Strength",
      "class_type": "Strength",
      "current_capacity": 10,
      "difficulty_level": "beginner",
      "duration_minutes": 60,
      "end_time": "11:00:00",
      "max_capacity": 10,
      "room_number": "Studio C",
      "schedule_date": "2026-05-13",
      "schedule_id": 19,
      "spots_available": 0,
      "start_time": "10:00:00",
      "status": "scheduled",
      "trainer_name": "Ali Rehman"
    }
  ],
  "success": true
}

```

T11 | Book without token → 401

Expected: 401 | Got: 401

```

{
  "error": "Authorization token is missing",
  "success": false
}

```

T12 | Non-member role → 403

Expected: 403 | Got: 403

```

{
  "error": "Access denied. Required role(s): m, e, m, b, e, r",
  "success": false,
  "your_role": "admin"
}

```

T13 | Booking: Morning Yoga on 2026-03-13 (15 spots)

T13 | Confirmed booking → 201

Expected: 201 | Got: 201

```

{
  "booking": {
    "booking_id": 54,
    "class_name": "Morning Yoga",
    "position_in_waitlist": null,

```

```
T13 | Confirmed booking → 201
  Expected: 201 | Got: 201
  {
    "booking": {
      "booking_id": 54,
      "class_name": "Morning Yoga",
      "position_in_waitlist": null,
      "schedule_date": "2026-03-13",
      "spots_remaining": 14,
      "start_time": "06:00:00",
      "status": "confirmed"
    },
    "message": "Class booked successfully \u2014 confirmed!",
    "success": true
  }
  Booking ID : 54
  Status      : confirmed
  Spots left   : 14

T14 | Duplicate booking → 409
  Expected: 409 | Got: 409
  {
    "error": "You have already booked this class",
    "existing_status": "confirmed",
    "success": false
  }

T15 | Full class: Beginner Strength (0 spots)

T15 | Full class → waitlist → 201
  Expected: 201 | Got: 201
  {
    "booking": {
      "booking_id": 55,
      "class_name": "Beginner Strength",
      "position_in_waitlist": 1,
      "schedule_date": "2026-05-13",
      "start_time": "10:00:00",
      "status": "waitlist"
    },
    "message": "Class is full. Added to waitlist at position 1.",
    "success": true
  }
  Waitlist position: 1

T16 | Invalid schedule_id=99999 → 404
  Expected: 404 | Got: 404
  {
    "error": "Schedule 99999 not found",
    "success": false
  }

T17 | Missing schedule_id → 400
  Expected: 400 | Got: 400
  {
    "error": "No data provided",
```

```

T17 | Missing schedule_id → 400
    Expected: 400 | Got: 400
    {
      "error": "No data provided",
      "success": false
    }

T18 | Cancel booking → 200
    Expected: 200 | Got: 200
    {
      "cancelled_booking_id": 54,
      "message": "Booking cancelled successfully",
      "note": "If confirmed, first waitlist member was automatically promoted",
      "success": true
    }

T19 | Cancel already cancelled → 400
    Expected: 400 | Got: 400
    {
      "error": "Booking is already cancelled",
      "success": false
    }

```

=====

EXPECTED RESULTS SUMMARY

=====

SCENARIO 1 - Membership Purchase:

```

T01: 400 – missing tier_id (validation)
T02: 400 – invalid payment method (validation)
T03: 404 – tier not found (transaction rolled back)
T04: 401 – no token (auth)
T05: 409 – already has membership (consistency)
T06: 201 – membership purchased (ACID commit)
T07: 409 – duplicate blocked (consistency)
T08: 200 – payment history

```

SCENARIO 2 - Class Booking:

```

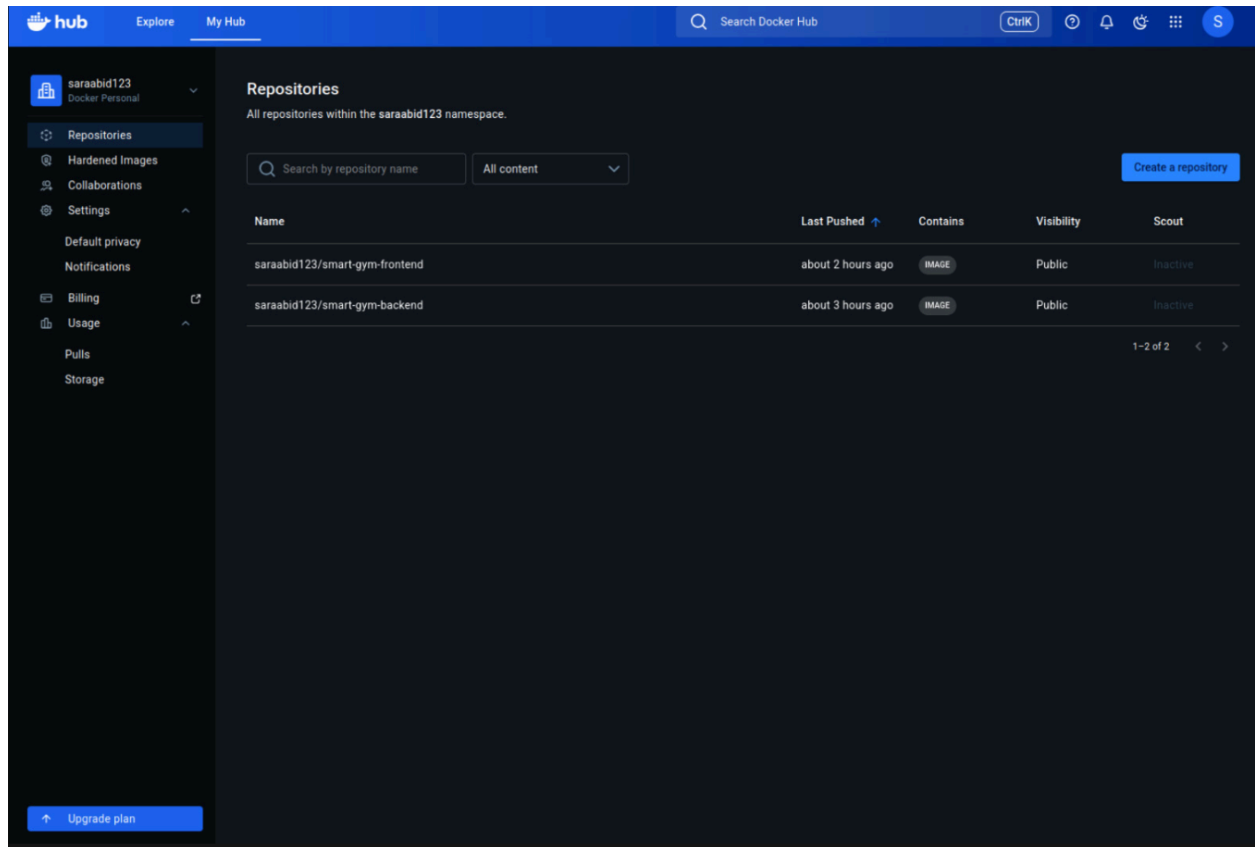
T09: 200 – public classes list
T10: 200 – public schedule
T11: 401 – no token
T12: 403 – wrong role
T13: 201 – confirmed booking (SELECT FOR UPDATE)
T14: 409 – duplicate booking blocked
T15: 201 – waitlist position (class full)
T16: 404 – invalid schedule
T17: 400 – missing schedule_id
T18: 200 – cancellation (trigger promotes waitlist)
T19: 400 – already cancelled

```

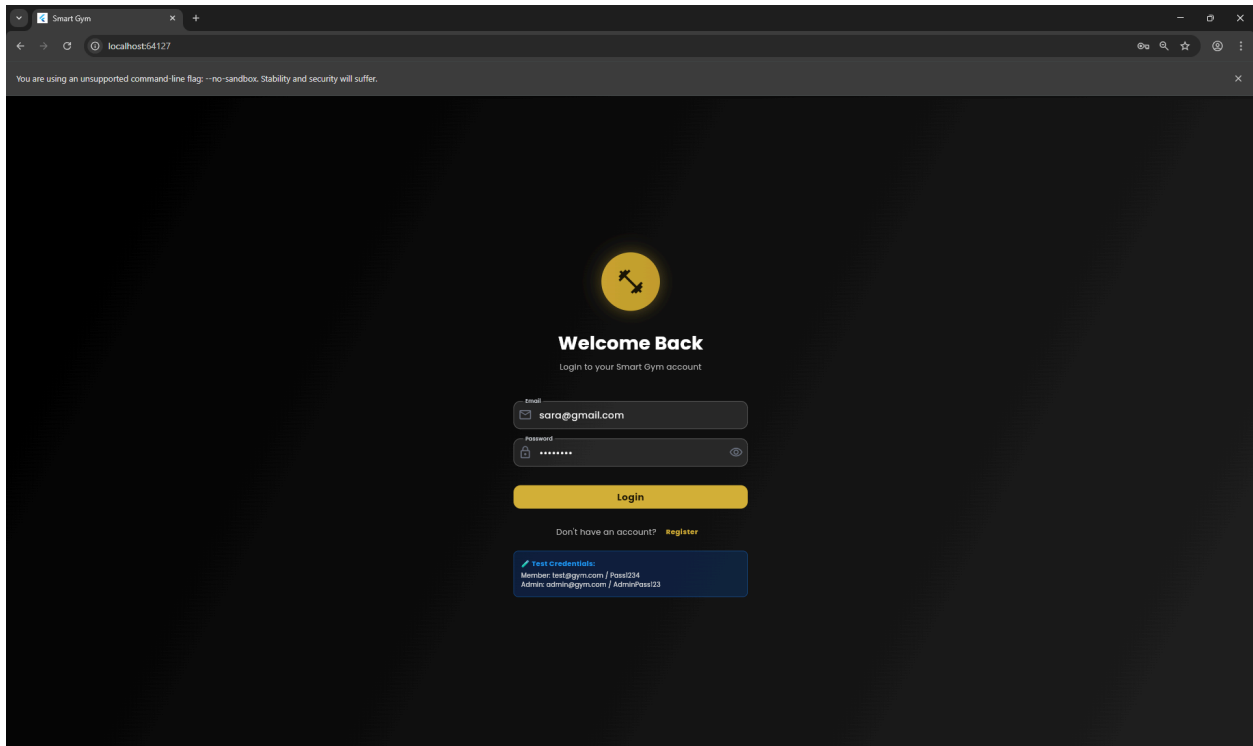
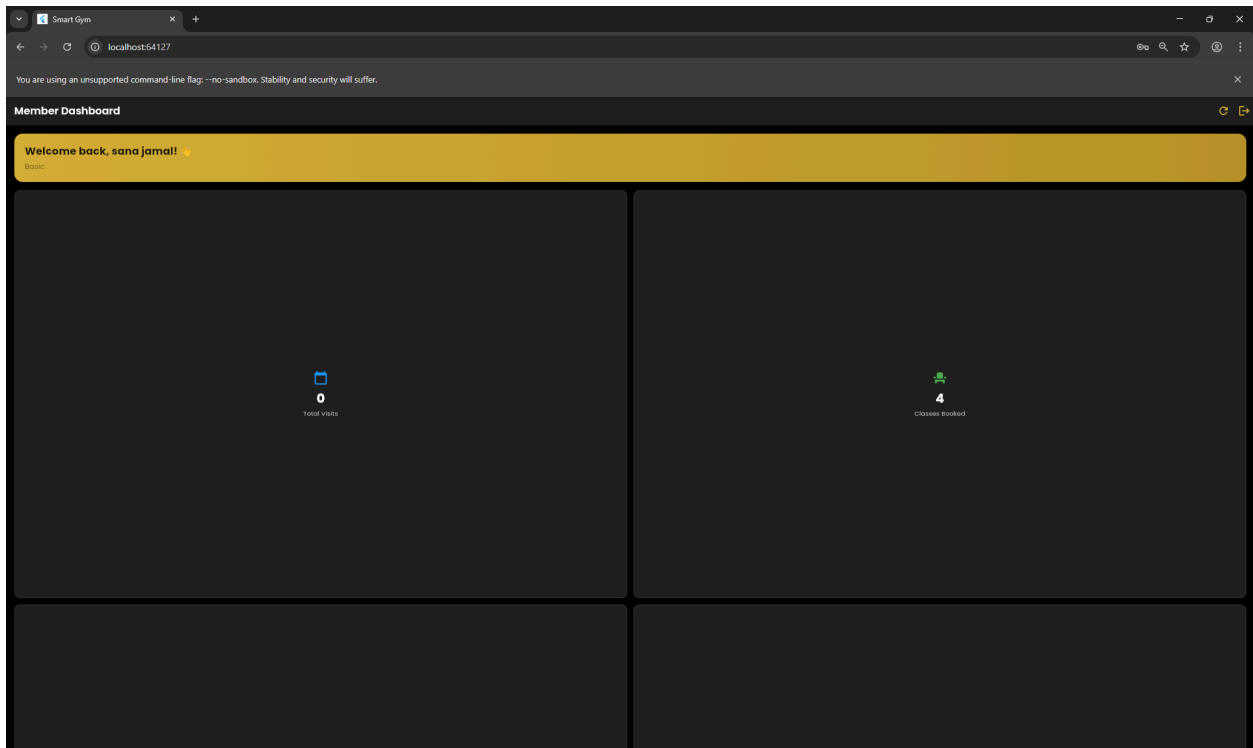
DEPLOYMENT :

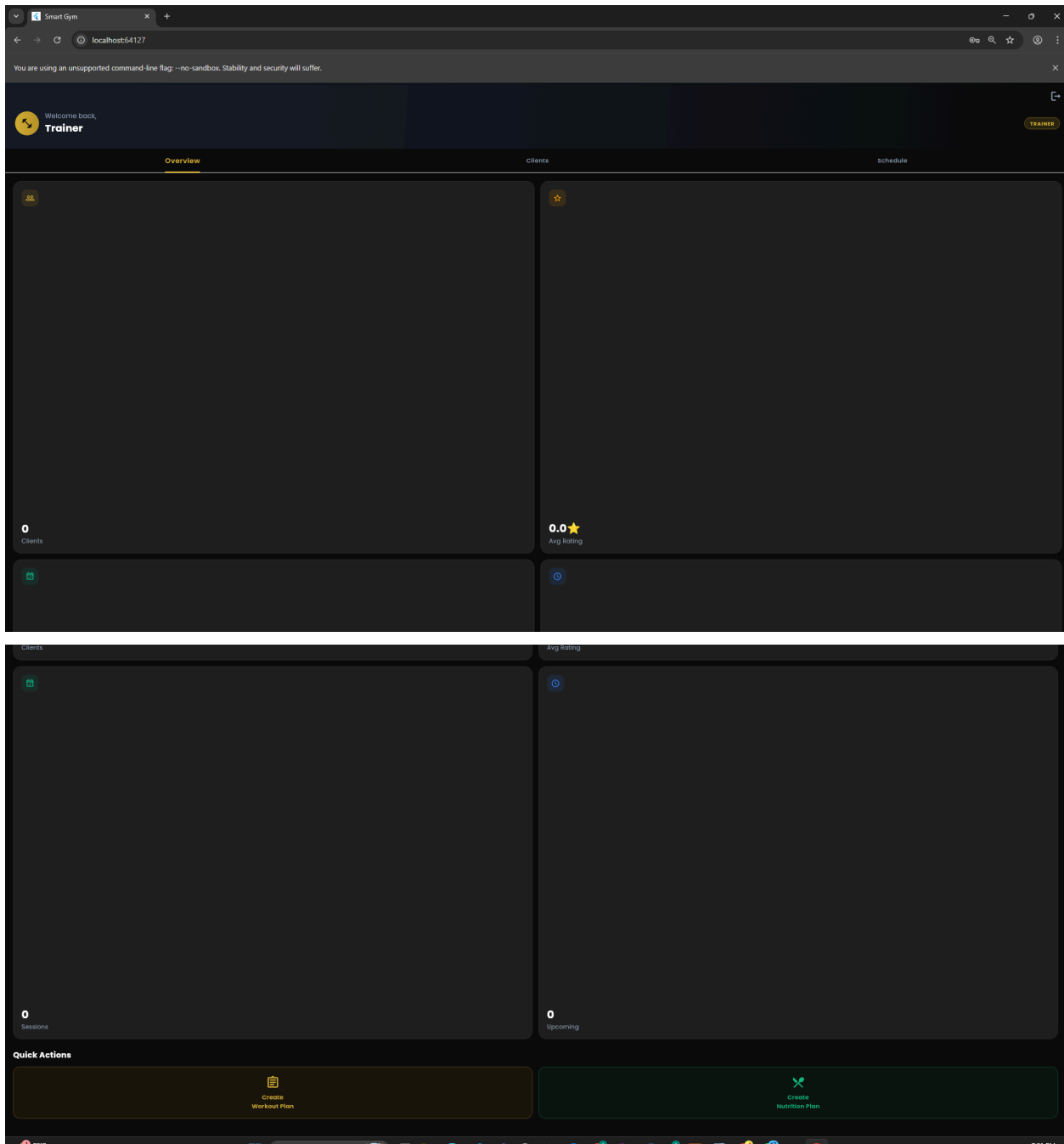
```
sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management$ docker compose up -d
WARN[0000] /home/sara-abid/smart_gym_management/docker-compose.yml: the attribute 'version' is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 3/3
  ✓ Container smart_gym_db      Healthy      0.6s
  ✓ Container smart_gym_frontend Started     0.9s
  ✓ Container smart_gym_backend Started     0.0s
sara-abid@sara-abid-ThinkBook-14-G6-IRL:~/smart_gym_management$
```

The screenshot displays the AWS Management Console interface for EC2 instances. The top navigation bar includes the AWS logo, search functionality, user profile (Sara Abid), and regional settings (Europe (Stockholm)). The left-hand sidebar contains various service categories: EC2 (selected), Dashboard, AWS Global View, Events, Instances (expanded), Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area is titled "Instances (1/1) Info". It features a table listing one instance named "smart-gym-se..." with ID "i-0a671c763b6e635f5", which is currently in a "Running" state. Below the table, there are tabs for "Details", "Status and alarms", "Monitoring", "Security", "Networking", "Storage", and "Tags". The "Details" tab is active, showing an "Instance summary" section with fields like Instance ID, IP addresses, Hostname type, Answer private resource DNS name, and Auto-assigned IP address. To the right of the summary, three columns provide additional details: Public IPv4 address (16.171.52.29), Instance state (Running), Private IP DNS name (ip-172-31-23-14.eu-north-1.compute.internal), Instance type (t3.micro), VPC ID (vpc-007e5aa10019b786), Private IPv4 addresses (172.31.23.14), Public DNS (ec2-16-171-52-29.eu-north-1.compute.amazonaws.com), Elastic IP addresses (none), and AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations).



FRONTEND:





Smart Gym

localhost:64127

You are using an unsupported command-line flag: --no-sandbox. Stability and security will suffer.

←

Create Account

Join Smart Gym today

First Name

Last Name

Email

Phone (optional)

Password

Confirm Password

Role

Member

Gender (optional)

Prefer not to say

Register

Already have an account? [Login](#)

←

→

Not secure 16.171.52.29:8080

🔍

🌐

🔒

🔧

🔗

👤

Welcome back,

Staff

STAFF

Check in

Attendance

Equipment

➡

1

Today's Check-ins

👤

1

Currently Inside

➡

Member Check-In

👤

Enter Member ID

➡

Check In

⬅

Check Out

Recent Check-Ins

👤

Member

2026-04-05 17:22:02 448889

Welcome back,
Staff

STAFF

Check InAttendanceEquipment

?

Member #null

© 2026-04-05 17:22:02.446889

Check Out

Welcome back,
Staff

STAFF

Check InAttendanceEquipment

✔

0

Operational

🔧

1

Maintenance

✖

0

Out of Order

✖

Elliptical 01

Out of Order

✖

Rowing Machine 01

Out of Order

✖

Stationary Bike 01

Out of Order

✖

Treadmill 01

Out of Order

✖

Treadmill 02

Out of Order

🔧

Treadmill 03

Maintenance

✖

Dumbbell Set

Out of Order

✖

Battle Ropes

Out of Order

✖

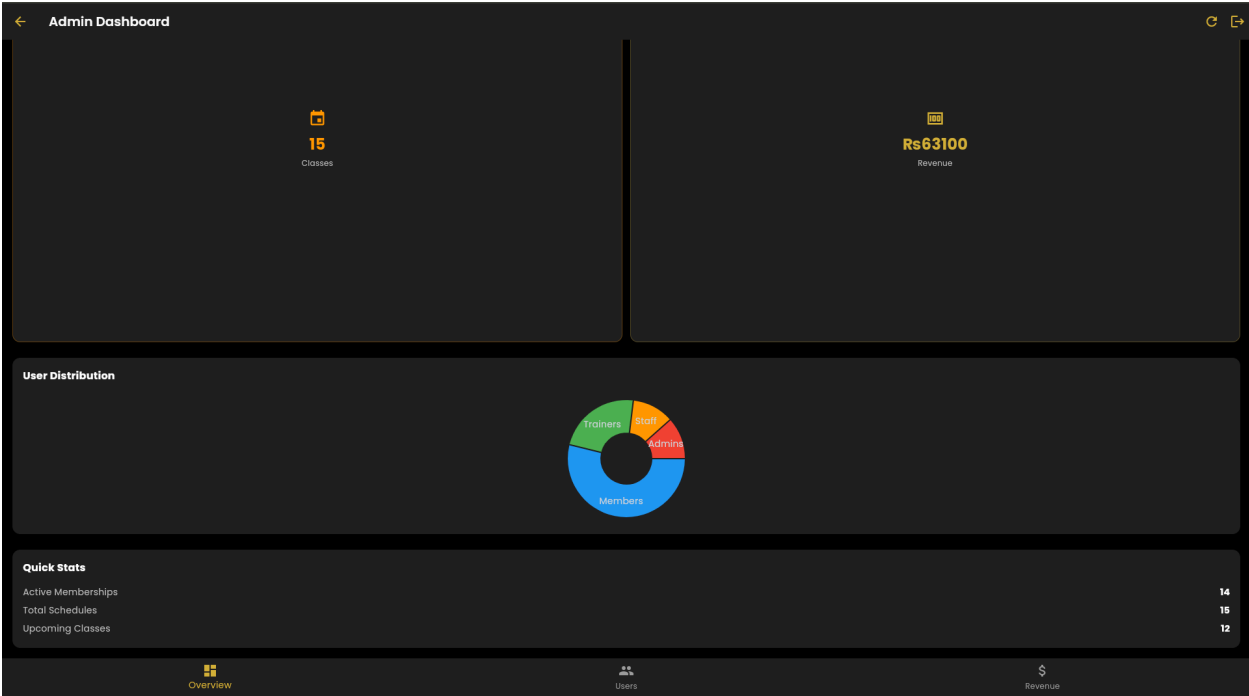
Kettlebell Set

Out of Order

✖

Pull-up Bar Station

Out of Order



Admin Dashboard

ak	admin khan adminkhan@gmail.com	admin	✓
ak	ammara khan ammara@gmail.com	staff	✓
mm	member member member@gmail.com	member	✓
aa	admin admin admin@gmail.com	admin	✓
RS	Rabia Syed rabiasyed3578@gmail.com	trainer	✓
sa	sara abid sara@gmail.com	member	✓
HR	Hamza Raza hamzaraza@gmail.com	member	✓
ZH	Zainab Hassan zainab.hassan@gmail.com	member	✓
BA	Bilal Ahmed bilal.ahmed@gmail.com	member	✓
MF	Maria Ferooz maria.feroaz@gmail.com	member	✓

